Mouhssine El Boumshouli

AI Engineering Student | Full-Stack & AI Software Developer | Founder, Convoroute LLC

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Born 28 November 2004, Fès, Morocco | Moroccan national

Profile

Fourth-year AI engineering student at EIDIA, Euromed University of Fes (graduating 2028), and founder of Convoroute LLC; I build full-stack products and reproducible AI research.

Education

EIDIA, Euromed University of Fes (UEMF) — Fès, Morocco
Engineering Degree in Artificial Intelligence, 5-year programme — *2023 – Expected 2028*

- Currently in the 4th year of the programme, which is the 2nd year of the engineering cycle; the integrated preparatory cycle (CPI) is completed.
- **Relevant coursework:** supervised machine learning, deep learning, computer vision, big data analytics, advanced databases, web and mobile development.

Institut Mazarin — Fès, Morocco
Baccalauréat, Physical Sciences, French option (international track) — *June 2023*

Research Experience

DARE-Bench – AI Agent Reliability Study — 2026

- A study of one question: do extra turns make an AI agent more reliable, or just more random?
- Raised measured AI-agent success from **34.2% to 50.0%** (+15.8 points) by giving the agent 5 interaction turns instead of 3, over **240 reproducible runs across 24 tasks**, 5 repetitions per condition.
- Made the finding trustworthy (flakiness also increased) with official rescoring of every run and 10,000-resample bootstrap confidence intervals. Code published openly (Apache 2.0).

medskel – Medical Image Skeletonization — 2026

- Software that turns vessel-like shapes in medical images into clean centrelines, so vessel length is measured the same way every time.
- Cut disagreement between two human observers on vessel-length measurement from **9.3% to 3.2%** across **28 retinal images** (lower on 26 of 28 images, $p < 0.0001$) by implementing a bisector-based skeletonization method and testing it against standard pixel thinning. Code published openly (MIT).
- Proved it with 28 hand-derived tests and 9 repeatable experiments, and reported the cost openly: about 66× slower runtime than thinning.

Professional Experience

Bounaim Auto — Beni Mellal, Morocco
Software & AI Engineering Intern — *July 1 – August 26, 2026*

- A platform for a car-import company: it compares supplier quotations and shows the true total cost of each offer, so the buyer sees the full picture.
- Centralised **5 procurement workflows** (supplier quotes, currency and cost analysis, document imports, AI-assisted data capture, human review) into one platform, SmartImport, built with React, TypeScript, FastAPI, PostgreSQL, and Docker.
- Reached **98.99% document-field F1** and **100% product-line F1** on synthetic evaluation, and **92.31% top-1 supplier matching** on eligible unique-match cases, by combining deterministic landed-cost logic with human-reviewed AI extraction.
- Verified with **139 automated backend tests**; incomplete offers are flagged and every analysis is kept on record.

Convoroute LLC — Remote
Founder & Software Developer — *June 2026 – Present*

- My own company: AI chatbots that plug into a business's website and answer its customers automatically.
- Founded Convoroute LLC and launched a B2B SaaS product covering **3 business needs** (customer support, FAQ automation, lead capture) by building AI chatbots that embed directly on client websites.
- Run the whole product myself – architecture, LLM integration, deployment, onboarding, and Stripe billing – in one repeatable workflow.

Projects

Recall – Multilingual Memory & Transcription App (React Native, Expo, SQLite, Gemini) 2026

- A mobile app that records conversations, classes, and meetings, transcribes them live in several languages, and turns them into clear notes, so nothing important is lost.
- Kept recordings safe through connection drops by storing a local WAV file while streaming audio in roughly 100 ms chunks for live transcription (sessions rotate about every 8.5 minutes).
- Turned transcripts into summaries, key points, action items, and chapters, stored in versioned SQLite with offline history and playback.

UEMF Présence – Smart Attendance System (Next.js, TypeScript, Prisma) 2026

- A digital attendance system for my university: it replaces paper sign-in sheets in class.
- Reduced false check-ins for **3 user groups** (students, professors, administrators) by verifying GPS location at check-in and refreshing HMAC-SHA256 QR codes every 10 seconds.
- Cut routine admin work with recurring schedules, review tools, CSV reports, and an anomaly dashboard.

AutoStock AI – Inventory & Forecasting for Odoo (Python, React) 2026

- An application that connects to Odoo to help a business track its stock and forecast what it will need next.
- Completed and merged the **first 6 development gates** of the application by October 2026, each gate shipped as a small pull request with automated tests.
- Built with an AI coding agent; every change independently reviewed (code, tests, screenshots) before merging.

Startup Internship Hunter (Python, Playwright, SQLite) 2026

- A tool I built for my own internship search: it collects public information about startups and prepares human-checked outreach.
- Built an evidence-first research pipeline with hard safety limits (max 12 pages and 2 repositories per company, no automated logins or forms) to prepare human-verified internship outreach.

Additional projects 2025 – 2026

- Also built: **Intelligent Student Performance System** (Java, 2025; logistic regression written from scratch); **Modern Finance Tracker** (C++17, Qt, SQLite, 2025); **Jev Voice Assistant** (Python, 2026; voice commands on a decision-model API); **food-expiry iOS app** (TypeScript, in development).

Technical Skills

- **Languages:** Python, TypeScript, JavaScript, Java, C, C++, SQL.
- **AI & Data:** LLM agents, AI evaluation, OpenAI and Gemini APIs, Google Cloud Speech-to-Text, scikit-learn, pandas, NumPy, OpenCV.
- **Web & Backend:** React, Next.js, React Native, FastAPI, SQLAlchemy, Pydantic, Prisma, PostgreSQL, SQLite.
- **Tools & Practices:** Git/GitHub, Docker, Linux, pytest, Playwright, Vercel, Nginx, Qt, CMake, Stripe, WordPress/WooCommerce, Cloudflare.

Certifications & Languages

- **UEMF Language Center** – C1 English (Cambridge English Placement Test), 26 October 2026.
- **TOEFL iBT** – 94/120 (Reading 20, Listening 22, Speaking 25, Writing 27), December 2022.
- **EF SET English Certificate** – 68/100, CEFR C1, August 2022 | **EnglishScore (British Council)** – CEFR C1, 535, August 2022.
- **Google** – Foundations of UX Design (Coursera) and Digital Marketing certificates, August 2022.
- **Languages:** Arabic (native) | English (C1) | French (B1) | Spanish (beginner). **Activities:** MSIC 2026 Hackathon | Idea'Wave Ideation Programme | 20+ GitHub repositories | Fiverr freelance app development.

Academic References

Pr. Asmae Abadi – Associate Professor at Euromed University of Fes; Coordinator of the Artificial Intelligence Program and the Industry 4.0 Program. Recommendation letter provided (October 2026).
Pr. Loubna Ourabah – Professor of Computer Science and Artificial Intelligence at EIDIA, Euromed University of Fes; Coordinator of the Big Data Analytics Program. Recommendation letter provided (October 2026).
Pr. Ayoub El Baraka – Physics researcher with a PhD from Sidi Mohamed Ben Abdellah University; taught Point Mechanics and MATLAB/Simulink Programming at Euromed University of Fes. Recommendation letter provided.